University of Basel

File Encryption Project

Final Project Presentation

Simon Zeugin, Marvin Georges, Nicholas Berger, Frédéric Weyssow

Operating Systems Lecture Spring Semester 2026

June 11, 2026

Outline

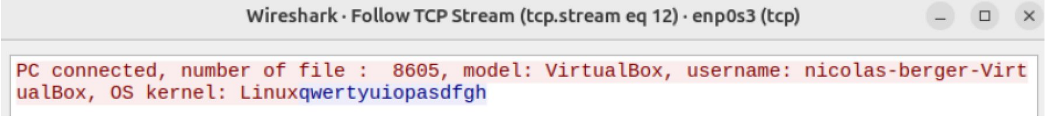
1. Problem Statement
2. Key Exchange
3. Multithreading
4. Encryption/OpenSSL
5. Watcher
6. Security Component
7. DEMO
8. Divide of work / Initial time
9. Lessons learned

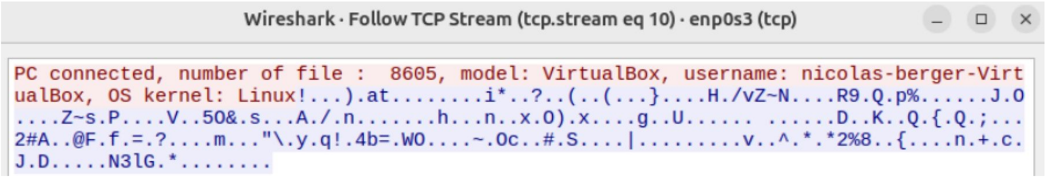
Problem Statement

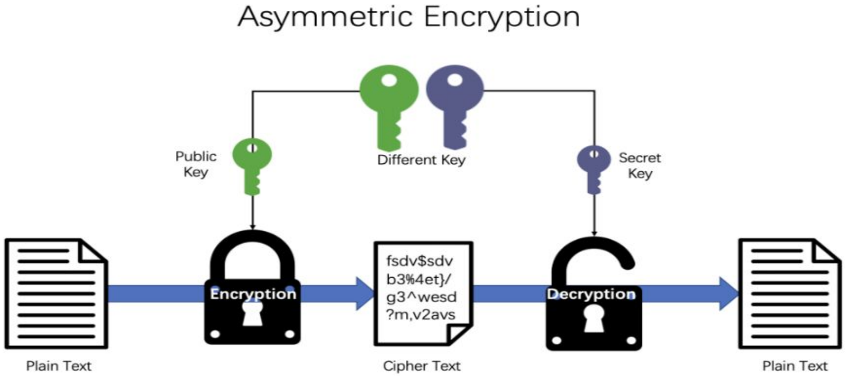
- Ransomware is one of the most destructive forms of malware
- How does ransomware actually work at the OS level and can we detect it in real time?
- Goal : Simulate a ransomware at the OS level, and build a real time defender.

Key exchange

- Simple Server/Client TCP connection
- Server holds the key and sends it to the client upon connection
- On the client side, key never persits in memory



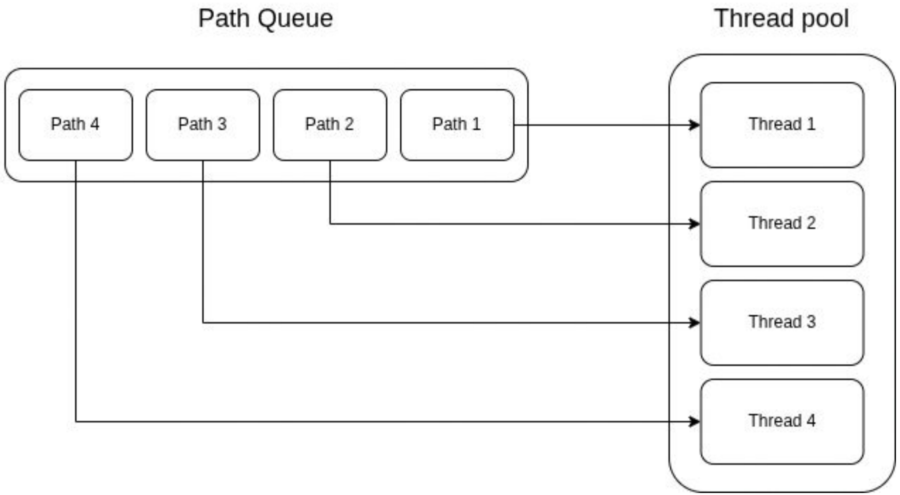




RSA-OAEP

Multithreading

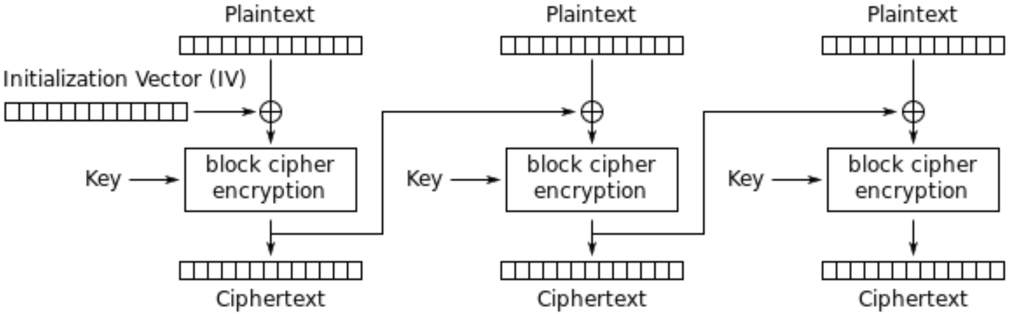
- **Why** do we need multithreading?
- **How** did we implement it?



Example with 4 threads

Encryption/Open SSL

- Original file is overwritten
- IV stored as a header in the encrypted file
- File renamed to .locked

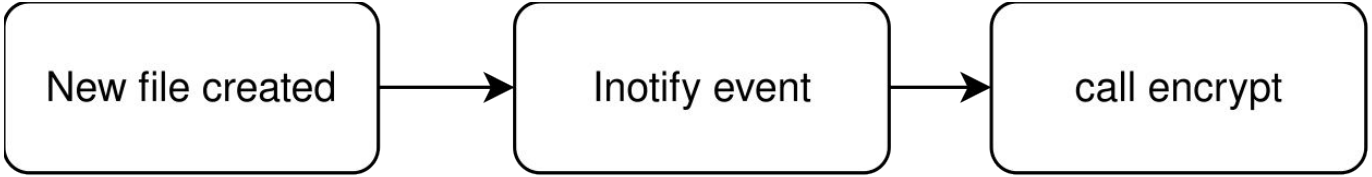


Cipher Block Chaining (CBC) mode encryption

AES-CBC-128

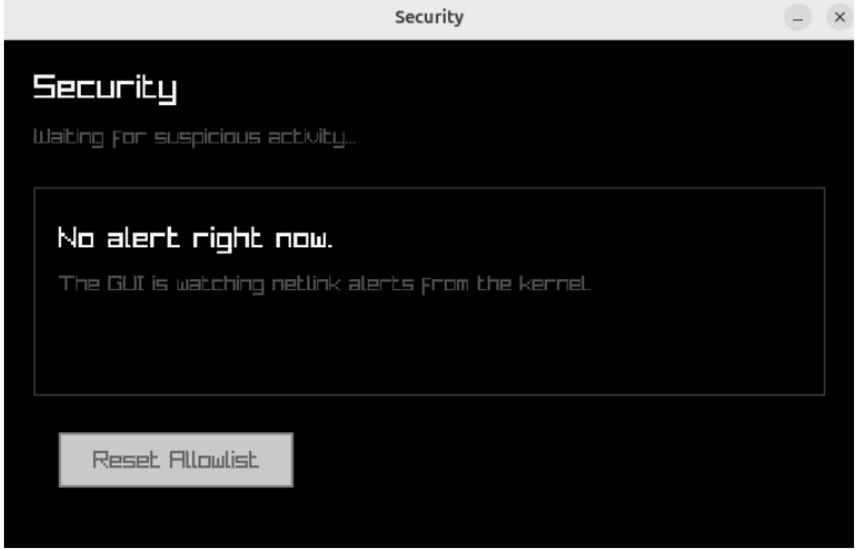
Watcher

- Why do we need a watcher?
- Encrypts new files
- Blocking future progress
- Inotify



Security Component

- Detects processes at kernel level
- Stops suspicious processes
- Gives the user the options: Kill, Trust or Continue

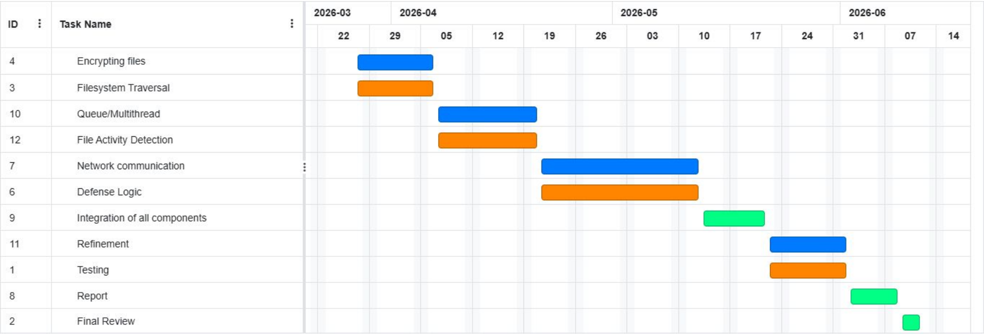


Demo

Division of work / Initial timeplan

- Frederic: Server/Client architecture, multithreading, key management, encryption, GUI.
- Simon: Multithreading, file queue, encryption, watcher, GUI.
- Nicolas: Encryption, Security Component, User/kernel space communication, GUI.
- Marvin: Encryption, Security Component, User/kernel space communication, GUI.

- Marvin George, Nicolas Berger
- Frédéric Weyssow, Simon Zeugin
- As a Team



Lessons Learned

- New technical skills:
  - Multithreading
  - Socket programming
  - OpenSSL
  - Inotify
  - Linux kernel module development with kprobes
- Good planning and time management
- Teamwork very important